Do Now · retrieval

Lesson 134 · Database Foundations: Flat Files, Tables and Keys

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a cryptographic digest help verify?

Q2

Which is removed first from a FIFO queue?

Q3

What is a composite key?

Q4

What are the outputs of a half adder when A=1 and B=1?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Single flat file — Duplicating pupil details on each enrolment creates repetition.

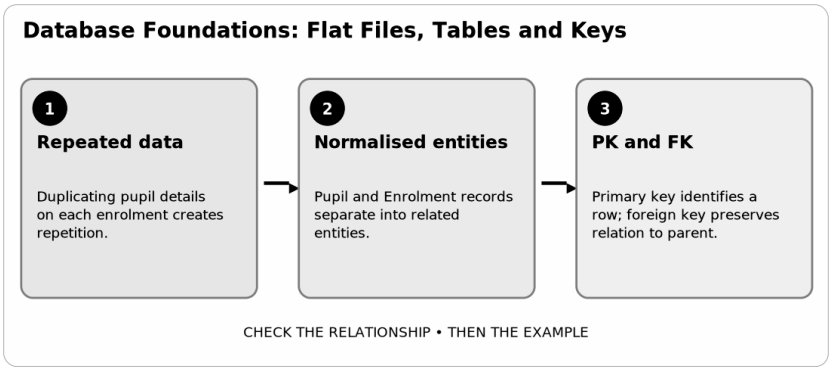
Core idea 2

Relational tables — Pupil and Enrolment records separate into related entities.

Core idea 3

Keys connect — Primary key identifies a row; foreign key preserves relation to parent.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

record — A row representing one entity instance.

primary key — Unique, non-null identifier for a table row.

foreign key — Column referencing a key in another table.

flat file — Data kept in one simple table or file without relational links.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A pupil books several clubs. Why store pupils and club bookings in separate tables?

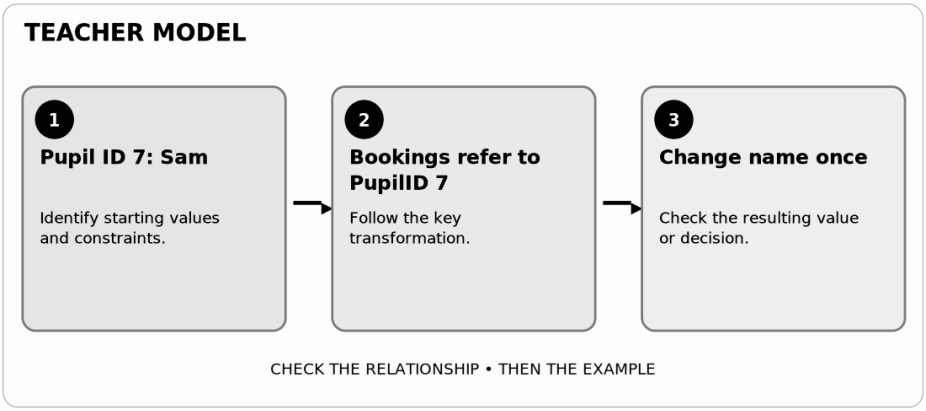
Follow the worked method

1. Pupil ID 7: Sam — Find the given data and name the required result.

2. Bookings refer to PupilID 7 — Explain why this intermediate stage is necessary.

3. Change name once — Pupil(PupilID PK, Name) and Booking(BookingID PK, PupilID FK, ClubID FK) separate data. A pupil name is held once and each booking refers to the pupil by ID, reducing update anomalies.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Device PK DeviceID
Find the given data and name the required result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define primary key and foreign key using pupil/club tables. [6]

Question 2

Explain two problems with keeping pupil contact details on every booking row. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Design a relational structure for Customers placing multiple Orders, showing suitable PKs and FK. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A foreign key value must always be unique in its table.

Use key vocabulary from memory

1. Explain record in your own words.

2. Explain primary key in your own words.

3. Explain foreign key in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.